

Exercise-2

Problem statement-Implementing the Factory Method Pattern

Source code:

```
package test;
```

```
import computer.Computer;
```

```
public class ComputerBuilderTest {
```

```
    public static void main(String[] args) {
```

```
        System.out.println("\n[1] Basic Office PC");
```

```
        Computer officePc = new Computer.Builder("Intel Core i3-13100")
```

```
            .ram("8GB DDR4")
```

```
            .storage("256GB SSD")
```

```
            .motherboard("ASUS Prime H610M")
```

```
            .powerSupply("450W Bronze PSU")
```

```
            .wifi(true)
```

```
            .build();
```

```
        System.out.println(officePc);
```

```
        System.out.println("\n[2] High-End Gaming Rig");
```

```
        Computer gamingRig = new Computer.Builder("AMD Ryzen 9 7950X")
```

```
            .ram("64GB DDR5-6000")
```

```
            .storage("4TB NVMe Gen5 SSD")
```

```

        .gpu("NVIDIA RTX 4090 24GB")
        .motherboard("ASUS ROG Crosshair X670E")
        .powerSupply("1000W Platinum PSU")
        .coolingSystem("360mm AIO Liquid Cooler")
        .wifi(true)
        .bluetooth(true)
        .build();

System.out.println(gamingRig);

System.out.println("\n[3] Budget Home Build");

Computer budgetBuild = new Computer.Builder("Intel Core i5-
12400")

        .ram("16GB DDR4")
        .storage("512GB SATA SSD")
        .powerSupply("550W Bronze PSU")
        .build();

System.out.println(budgetBuild);

System.out.println("\n[4] Professional Workstation");

Computer workstation = new Computer.Builder("Intel Xeon W-
2295")

        .ram("128GB ECC DDR4")
        .storage("8TB NVMe RAID")
        .gpu("NVIDIA RTX A6000 48GB")
        .motherboard("ASUS Pro WS W480-ACE")
        .powerSupply("1200W Titanium PSU")
        .coolingSystem("Custom Water Loop")

```

```

        .wifi(false)
        .bluetooth(false)
        .build();
System.out.println(workstation);

System.out.println("\n[5] Minimal Server Node (CPU only)");
Computer server = new Computer.Builder("AMD EPYC
7763").build();
System.out.println(server);
System.out.println("\n[6] Testing validation — missing CPU:");
try {
    Computer invalid = new Computer.Builder("").build();
} catch (IllegalArgumentException e) {
    System.out.println(" Caught expected error: " +
e.getMessage());
}
System.out.println("\nAll tests passed successfully.\n");
}
}

```

Output:

Problems @ Javadoc Declaration Console × Terminal

<terminated> DocumentFactoryTest [Java Application] C:\Users\saich\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full:

Factory Method Pattern - Document Test

--- Test 1: Creating Documents via Factory

Created: Word Document

Created: PDF Document

Created: Excel Document

Result: PASSED

--- Test 2: Processing Each Document

Processing: Word Document

[Word] Opening Word document (.docx)

[Word] Saving Word document (.docx)

[Word] Closing Word document (.docx)

Processing: PDF Document

[PDF] Opening PDF document (.pdf)

[PDF] Saving PDF document (.pdf)

[PDF] Closing PDF document (.pdf)

Processing: Excel Document

[Excel] Opening Excel document (.xlsx)

[Excel] Saving Excel document (.xlsx)

[Excel] Closing Excel document (.xlsx)

Result: PASSED

--- Test 3: Document Type Verification

Word type correct? true

PDF type correct? true

Excel type correct? true

Result: PASSED

All Factory Method tests completed.